

APPENDIX C

SERVICEABILITY CONSIDERATIONS

C.1 SERVICEABILITY CONSIDERATIONS

This appendix is not a mandatory part of the standard but provides guidance for design for serviceability to maintain the function of a building and the comfort of its occupants during normal usage. Serviceability limits (e.g., maximum static deformations or accelerations) shall be chosen with due regard to the intended function of the structure.

Serviceability shall be checked using appropriate loads for the limit state being considered.

C.2 DEFLECTION, DRIFT, AND VIBRATION

C.2.1 Vertical Deflections. Deformations of floor and roof members and systems caused by service loads shall not impair the serviceability of the structure.

C.2.2 Drift of Walls and Frames. Lateral deflection or drift of structures and deformation of horizontal diaphragms and bracing systems caused by wind effects shall not impair the serviceability of the structure.

C.2.3 Vibrations. Floor systems supporting large open areas free of partitions or other sources of damping, where vibration caused by pedestrian traffic might be objectionable, shall be designed with due regard for such vibration.

Mechanical equipment that can produce objectionable vibrations in any portion of an inhabited structure shall be isolated to minimize the transmission of such vibrations to the structure.

Building structural systems shall be designed so that wind-induced vibrations do not cause occupant discomfort or damage to the building, its appurtenances, or contents.

C.3 DESIGN FOR LONG-TERM DEFLECTION

Where required for acceptable building performance, members and systems shall be designed to accommodate long-term irreversible deflections under sustained load.

C.4 CAMBER

Special camber requirements that are necessary to bring a loaded member into proper relation with the work of other trades shall be set forth in the design documents.

Beams detailed without specified camber shall be positioned during erection so that any minor camber is upward. If camber involves the erection of any member under preload, this shall be noted in the design documents.

C.5 EXPANSION AND CONTRACTION

Dimensional changes in a structure and its elements caused by variations in temperature, relative humidity, or other effects shall not impair the serviceability of the structure.

Provision shall be made either to control crack widths or to limit cracking by providing relief joints.

C.6 DURABILITY

Buildings and other structures shall be designed to tolerate long-term environmental effects or shall be protected against such effects.

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